Machine Learning *and* Smartphones: A Powerful Combination

The combination of smartphones and machine learning presents a formidable tool to solve problems that were almost impossible to overcome earlier. As mobile hardware becomes more powerful, mobile computing abilities will now cross new thresholds.

The paradigm shift in programming—from a sequential instruction based approach to a learning based approach—has offered new opportunities to solve problems which were hitherto almost impossible to address. Considering the ubiquitous nature of smartphones, it becomes mandatory to port machine learning (ML) to smartphone environments. This article focuses on this important and powerful combination of machine learning and mobile devices.

The traditional programming approach requires the programmer to incorporate explicit instructions on how to respond in each and every scenario. Though the certainty of how the program flows can be confirmed with 100 per cent accuracy in the traditional approach, a major pitfall is the lack of support when it comes to responding to novel scenarios. Unfortunately, most of today's problems require the ability to cope with novel scenarios. For example, it would be nearly impossible to build a writer-independent handwriting recognition system with maximum accuracy using the traditional sequential instruction based approach. However, if you shift towards the learning based approach, such problems can be handled in an efficient manner. The recent widespread use of machine learning has certainly enabled programmers to build solutions such as speaker-independent voice recognition, image understanding, distortion-resistant face recognition, etc.

Many of the problems mentioned here require solutions to be portable, i.e., they will be very useful if they run on smartphones. Hence, combining the power of machine learning and the portability of smartphones makes for a promising model in providing solutions to complex problems.

Machine learning and deep learning

Machine learning and its recent evolution, deep learning, have been explored by many researchers in recent times. There are many frameworks and libraries available to assist programmers in implementing ML based approaches. Some of them are listed in Figure 1.

Each of these frameworks offers certain specialised services.